No.

Date . .

Identities and law

1. Common factors

$ab + ac = a(b+c)$

$ab - ac = a(b-c)$

$-ab - ac = -a(b+c)$

usually for parameter or area

$P = 2w + 2L$

$80 = 2w + 2L$ ÷2

$40 = w + L$

remember when do the factors / eq.

① Simplify by ÷ or x

÷ if same factors

x if it fraction → whole number

② only multiple that with

2. Perfect square identities

$a^2 + 2ab + b^2 = (a+b)^2$

$a^2 - 2ab + b^2 = (a-b)^2$

3. Difference of two squares

$a^2 - b^2 = (a+b)(a-b)$

" shows it it same

$A = x(40 - x)$ "

Find w and L from $40 = w + L$

if $w = x$

(diff var) $\frac{1}{3}x + 2y = 10$

$(3)\frac{1}{3}x + (3)2y = (3)10$

(X) $3y + \frac{1}{3}y + 2x = 10$

$3\frac{1}{3}y + 2x = 10$

$\frac{9+1}{3}y + 2x = 10$

$\frac{10}{3}y + 2x = 10$

Next step : after do ↑

2. zero product principle

$a(b-c) = 0$

$a = 0$ $b - c = 0$

$(a+b)(a-c)$

$a+b=0$ $a-c=0$

$a = -b$ $a = +c$

ⓘ $40 = w + L$

$40 = x + L$

$(40 - L) = x$

ⓘⓘ $A = w \cdot L$

$A = x(40 - L)$ ≈ Question

③ Find factors by same common factors

or using :

$a = \square$

$b = \square$

$c = \square$

$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$a(b+c) = d$

ex.

$3x(x+1) = -3$

solve id $a(b+c) = ab + ac$ x $+1$

$3x$ $2x^2$ $3x$

move -3 $3x^2 + 3x = -3$

do zero product. $3x^2 + 3x - 3 = 0$

No.

Date

1. Solve each equations

common factor (cf)

id: $a(b-c) = ab - ac$

a -a

id: zero product principle

$a(b-c) = 0$

$a = 0 \quad b - c = 0$

(a) $a(a-9) = 0$

id:z $a = 0 \quad a - 9 = 0$

$a = 9$

$a = 0$ or 9

(b) $b(b+7) = 0$

id:z $b = 0 \quad b + 7 = 0$

$b = -7$

$b = 0$ or -7

(c) $5c^2 + 25c = 0$

id:cf $5c^2 + 25c = 0$ ÷ 5 c

$5 \cdot c \quad 5 \cdot 5 \cdot c$

id:z $5c(c+5) = 0$

$c = 0 \quad c + 5 = 0$

$c = -5$

$c = 0$ or -5

(d) $11d^2 - d = 0$

id:cf $11d^2 - d = 0$ ÷ d

$11 \cdot d \cdot d \quad -d$

id:z $d(11d - 1) = 0$

$d = 0 \quad 11d - 1 = 0$

$d = \frac{1}{11}$

$d = 0$ or $\frac{1}{11}$

(e) $-4h^2 - 16h = 0$

cf $-4h^2 - 16h = 0$ ÷ 4 h

$-4 \cdot h \cdot h \quad 4 \cdot 4 \cdot h$

z $-4h(h + 4) = 0$

$h = 0 \quad h + 4 = 0$

$h = -4$

$h = 0$ or $h = -4$

(f) $3k - 81k^2 = 0$

reorder: $ax^2 + bx + c = 0$

reorder cf $-81k^2 + 3k = 0$ ÷ 3 k

z $-3k(27k - 1) = 0$

$k = 0 \quad 27k - 1 = 0$

$k = \frac{1}{27}$

$k = 0$ or $\frac{1}{27}$

(g) $-\frac{1}{2}x(2x + 3) = 0$

$-\frac{1}{2}x = 0 \quad 2x + 3 = 0$

$2x = -3$

$x = -\frac{3}{2}$

$x = 0$ or $-\frac{3}{2}$

(h) $\frac{4}{5}y^2 - 4y = 0$

$\frac{4}{5} \cdot y \cdot y \quad -4 \cdot 5 \cdot y$ ÷ $\frac{4}{5}$ y

$\frac{4}{5}y(y - 5) = 0$

$\frac{4}{5}y = 0 \quad y - 5 = 0$

$y = 0$ or 5

$y = 5$

No.

Date

2. Solve each of following eq using zero product principle

(a) $(m-4)(m-9)=0$

$m = 4$ or 9

(b) $(n-3)(n+5)=0$

$n = 3$ or -5

(c) $(p+1)(p+2)=0$

$p: -1$ or -2

(d) $(7q-6)(4q-5)=0$

$7q-6=0$ $\quad$ $4q-5=0$

$7q=6$ $\quad$ $4q=5$

$q=\frac{6}{7}$ $\quad$ $q=\frac{5}{4}$

$q=\frac{6}{7}$ or $1\frac{1}{4}$

(e) $(5s+3)(2-s)=0$

$s=-\frac{3}{5}$ or 2

(f) $(-2t-5)(8t-5)=0$

$-2t-5=0$ $\quad$ $8t-5=0$

$-2t=5$ $\quad$ $8t=5$

$t=-\frac{5}{2}$ $\quad$ $t=\frac{5}{8}$

$t=-\frac{5}{2}$ or $\frac{5}{8}$

$-2\frac{1}{2}$

3. Solve each of following eq

(a) $s^2+10s+21=0$

$(s+3)(s+7)$

② $s+3=0$ $\quad$ $s+7=0$

$s=-3$ $\quad$ $s=-7$

$s=-3$ or -7

3(b) $t^2-16t+63=0$

$(t-7)(t-9)$

$t-7=0$ $\quad$ $t-9=0$

$t \quad 7$ $\quad$ $t \quad 9$

3(c) $3u^2+49u+60=0$

$ac \neq 0$ you can factorise normally.

cheat way: 60

4	15
5	12

check ttal and diff

$(3x+4)(x+15)$ $\quad$ $3.15=45$, $4=4$, 49

$(3x+15)(x+4)$

$(3x+5)(x+12)$

$(3x+12)(x+5)$

$3u^2+49u+60=0$

② $(3u+4)(u+15)=0$

$3u+4=0$ $\quad$ $u+15=0$

$u=-\frac{4}{3}$ or -15

Remember:

1. x^2+x-6

$(x-2)(x+3)$

2. x^2-x-6

$(x+2)(x-3)$

3. x^2+9 : perfect square

$(\quad)$ $\quad$ $a^2-b^2=(a+b)(a-b)$

$(a+b)^2$ or $(a-b)^2$

KOKUYO LOOSE-LEAF ノ-816A 7 mm ruled×35 lines

No.

Date

3(d) $6w^2 - 29w + 20 = 0$

$(6w - 5)(w + 4)$

$6w - 5 = 0$ $\quad w + 4 = 0$

$6w = 5$ $\quad w = -4$

$w = \frac{5}{6}$

$w = \frac{5}{6}$ or -4

3(f) $9y^2 + 21y - 18 = 0$ $\div 3$

$3y^2 + 7y - 6 = 0$

$(3y - 2)(y + 3)$

$3y - 2 = 0$ $\quad y + 3 = 0$

$3y = 2$ $\quad y = -3$

$y = \frac{2}{3}$

$y = \frac{2}{3}$ or -3

(h) $k^2 + 12k + 36 = 0$

$2(6)(k)$ $\quad \sqrt{36} = 6$

$12k$

perfect square $(a+b)^2$

$\rightarrow k^2 + 12k + 36 = 0$

$(k+6)^2 = 0$

$(k+6)(k+6) = 0$

$k + 6 = 0$

$k = -6$ $\quad k = -6$

(i) $25p^2 + 70p + 49 = 0$

$\sqrt{25p^2}$ $\quad 2(5p)(7)$ $\quad \sqrt{49} = 7$

$5p$ perfect square $(a+b)^2$

$\rightarrow 25p^2 + 70p + 49 = 0$

$(5p + 7)^2 = 0$

$5p + 7 = 0$

$p = -\frac{7}{5}$ $\quad 5p = -7$

-1.4 $\quad p = \frac{-7}{5}$.

(e) $\frac{1}{3}x^2 + 2x - 9 = 0$

$\frac{1}{3}(\quad)(\quad)$ cannot factorise without simplify.

$(3)\frac{1}{3}x^2 + (3)2x - (3)9 = (3)0$

$x^2 + 6x - 27 = 0$

$(x + 9)(x - 3)$

$x + 9 = 0$ $\quad x - 3 = 0$

$x = -9$ $\quad x = 3$

$x = -9$ or 3

(g) $m^2 - 16m + 64 = 0$

$\downarrow$ $\quad \downarrow$ $\quad \downarrow$

a^2 $\quad 2ab$ $\quad b^2$

$2(m)(-8)$ $\quad \sqrt{64} = 8$

$16m$

? perfect square

$\rightarrow m^2 \ominus 16m + 64 = (m - 8)^2$

$\rightarrow$ zero product principle

$\rightarrow m^2 - 16m + 64 = 0$

$(m - 8)^2 = 0$

$(m - 8)(m - 8) = 0$

$m - 8 = 0$

$m = 8$ $\quad m = 8$

(j) $\frac{4}{9}q^2 - \frac{4}{3}q + 1 = 0$

3

$(9)\frac{4}{9}q^2 - (9)\frac{4}{3}q + (9)1 = (9)0$

$4q^2 - 3q + 9 = 0$

$(2q - 3)^2 = 0$

$2q = 3$

$q = \frac{3}{2}$

$q = \frac{3}{2}$

KOKUYO LOOSE-LEAF ノ-816A 7 mm ruled×35 lines

No.

Date

5. Solve each the following equation

1. Expand
2. Zero product principle
3. Fachorise ← Id → common factor ( )( )
   ↘ $x_{1,2}$
4. zero product principle

5(a) $k(2k+5) = 3$

	$2k$	$+5$
k	$2k^2$	$5k$

(e) $2k^2 + 5k = 3$

(t) $2k^2 + 5k - 3 = 0$

(f) $(2k-1)(k+3)$

(z) $2k - 1 = 0$ $\quad k + 3 = 0$

$2k = 1$ $\quad k = -3$

$k = \frac{1}{2}$

$k = \frac{1}{2}$ or -3

5(b) $2m(m-5) = 5m - 18$

	m	-5
$2m$	$2m^2$	$-10m$

$2m^2 - 10m = 5m - 18$

$-5m \quad +18$

$2m^2 - 10m - 5m + 18 = 0$

$2m^2 - 15m + 18 = 0$

$2m^2 \quad -15m \quad +18$

$(2m-3)(m-6) = 0$

18	
2	9
6	3

$2m - 3 = 0$ $\quad m - 6 = 0$

$2m = 3$ $\quad m = 6$

$m = \frac{3}{2}$ $\quad m = 6$ or $\frac{3}{2}$

5(c) $(n-2)(n+4) = 27$

id $(a+b)(a+c) = d$

$a = -2$ $\quad x = n$

$x^2 + (a+b)x + ab = d$

$b = +4$ $\quad$ using (+/−)

(id) $x^2 + (a+b)x + ab = d$

$n^2 + (-2+4)n + (-2)(4) = 27$

$n^2 + 2n - 8 = 27$

(f) $n^2 + 2n - 35 = 0$

$(n-5)(n+7)$ $\quad n = 5$ or -7

$n - 5 = 0$ $\quad n + 7 = 0$

$n = 5$ $\quad n = -7$

KOKUYO LOOSE-LEAF ノ-816A 7 mm ruled×35 lines

5(f) $(3s+1)(s-4) = -5(s-1)$

$3s^2 - 6s - 9 = 0$ ÷ 3

$s^2 - 2s - 3 = 0$

$(s-3)(s+1) = 0$

$s - 3 = 0$ $\quad$ $s + 1 = 0$

$s = 3$ $\quad$ $s = -1$

$s = 3$ or -1

6. The sum of whole number a
and
twice of the square of the number
2 (a^2)
is 10

Find the number

$\Rightarrow a + 2a^2 = 10$

reorder $2a^2 + a - 10 = 0$

factors $(2a+5)(a-2) = 0$

$2a + 5 = 0$ $\quad$ $a - 2 = 0$

$2a = -5$ $\quad$ $a = 2$

$a = -\frac{5}{2}$

variable check

If $a = -\frac{5}{2}$

$a + 2a^2 = 10$

$-\frac{5}{2} + 2\left(-\frac{5}{2}\right)^2 = 10$

$-\frac{5}{2} + 2\left(-\frac{5}{2}\right)\left(-\frac{5}{2}\right) = 10$

$\times \quad -\frac{5}{2} + 2\left(\frac{25}{4}\right) = 10$

$a = 2$

$a + 2a^2 = 10$

$2 + 2(2)^2 = 10$

$2 + 2.4 = 10$

$2 + 8 = 10$ ✓

number $\quad$ number = 2